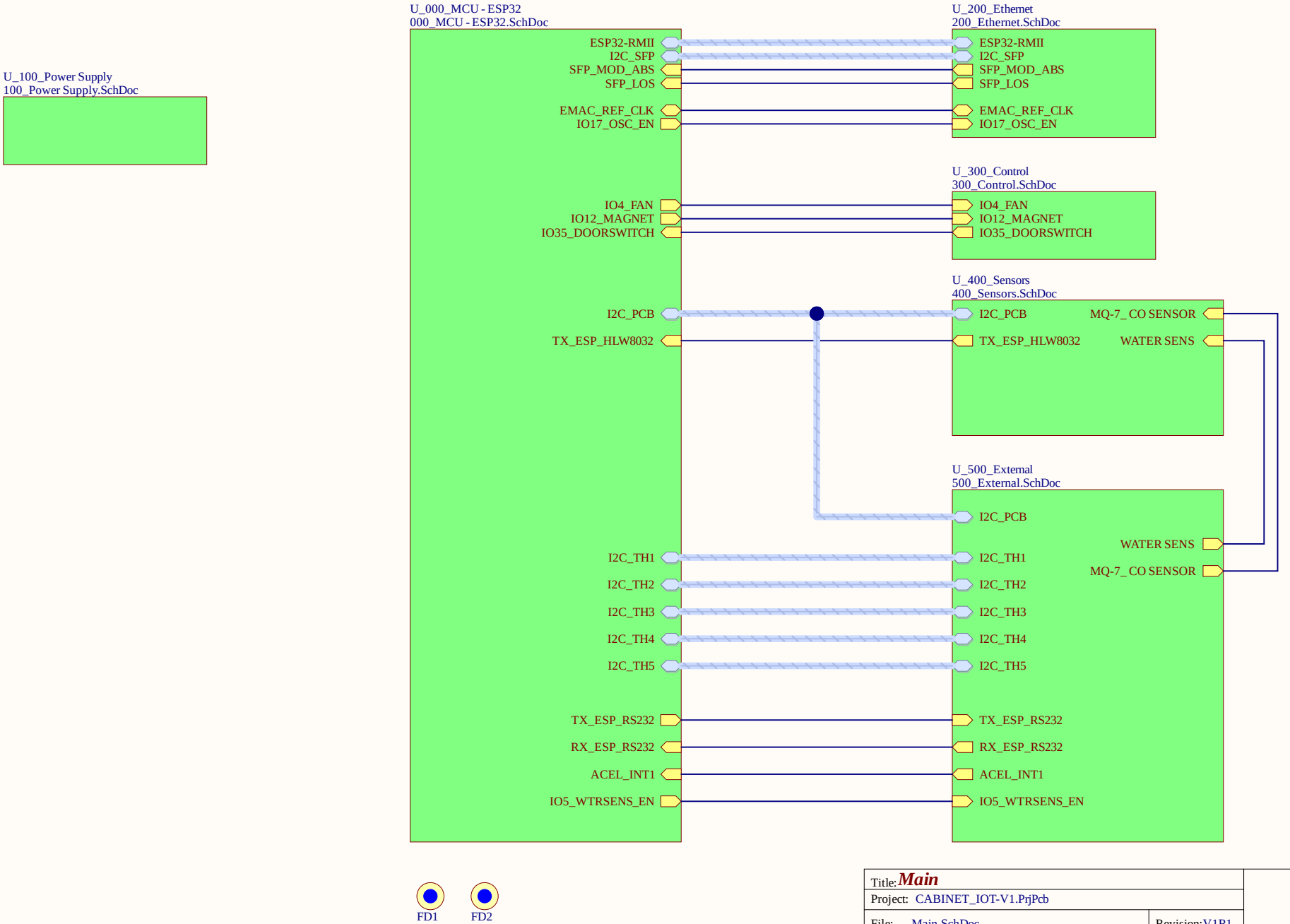
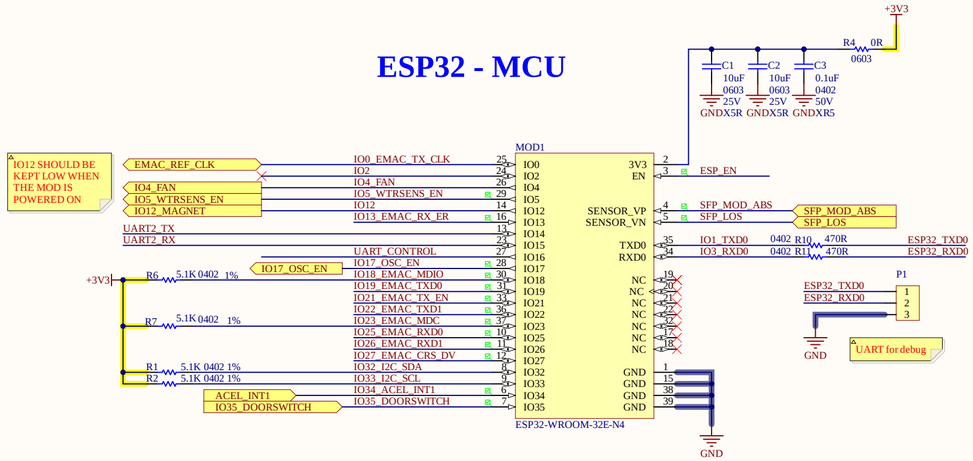
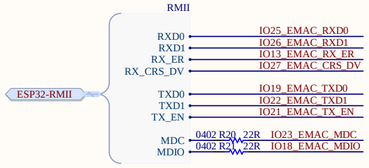


Title: ***Main***			
Project: CABINET_IOT-V1.PrjPcb			
File: Main.SchDoc		Revision: V1R1	
Date: 20261112	Size: A4	Drawn by: J-E.Pachon-J.Jaramillo	

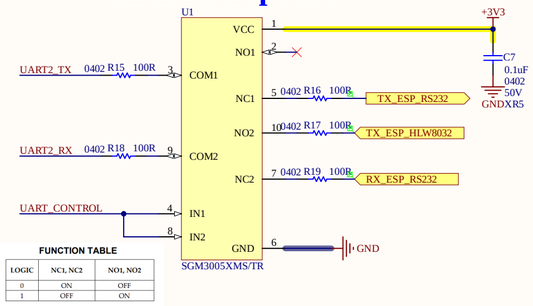
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ESP32 - MCU

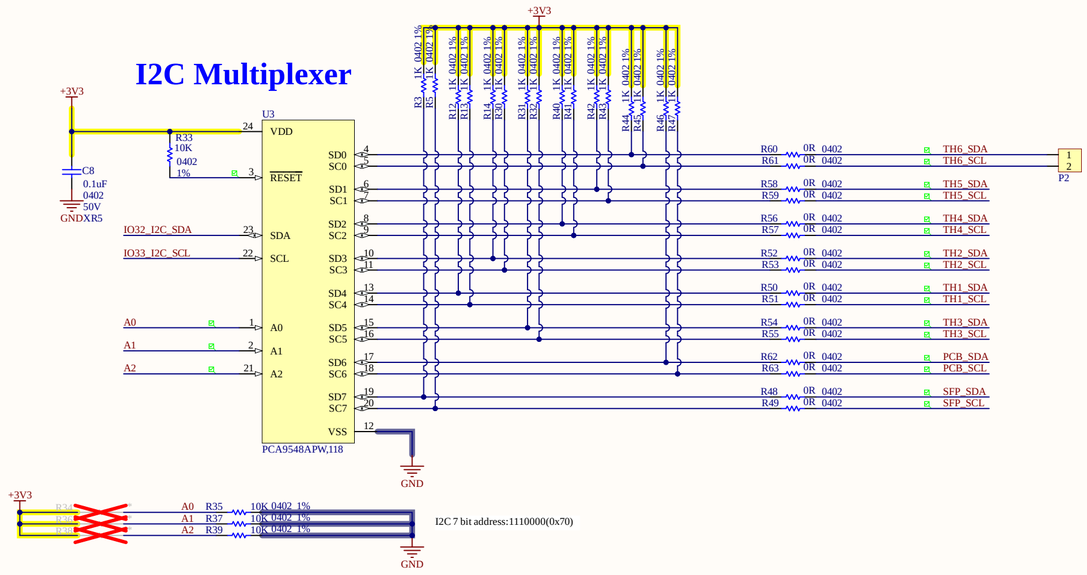


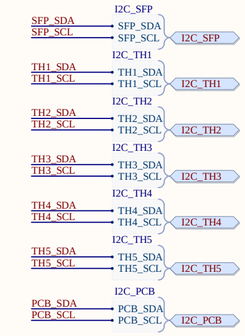


UART Multiplexer

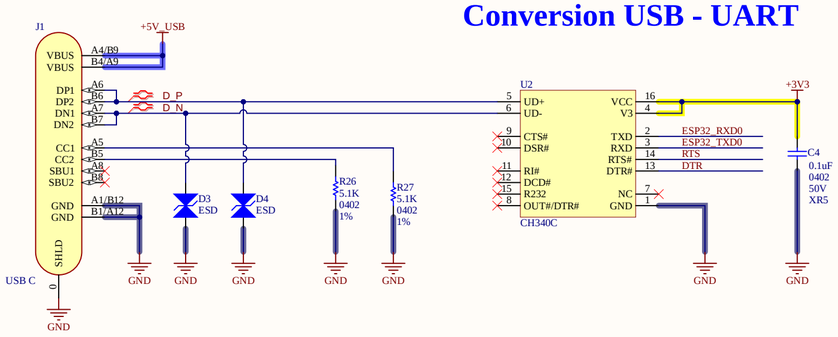


I2C Multiplexer

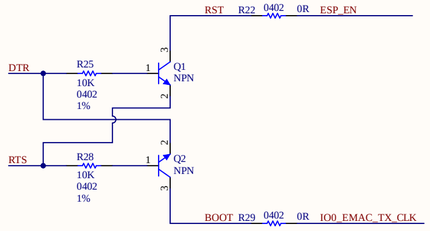




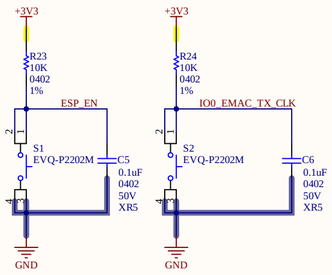
Conversion USB - UART



Auto Reset / Boot



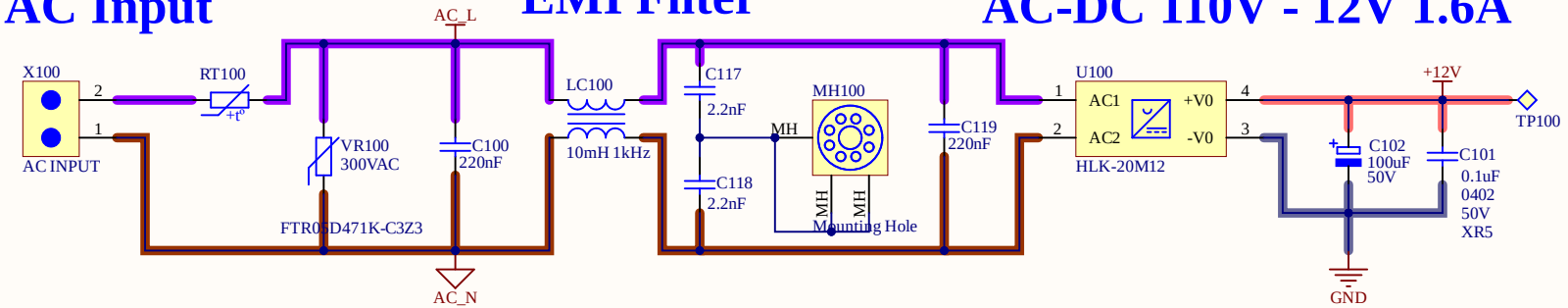
Buttons Reset / Boot



Title:	***MCU-ESP32***		
Project:	CABINET_IOT-V1.PrjPcb		
File:	000_MCU - ESP32.SchDoc	Revision:	V1R1
Date:	20261112	Size: A2	Drawn by: J-E.Pachon-J.Jaramillo

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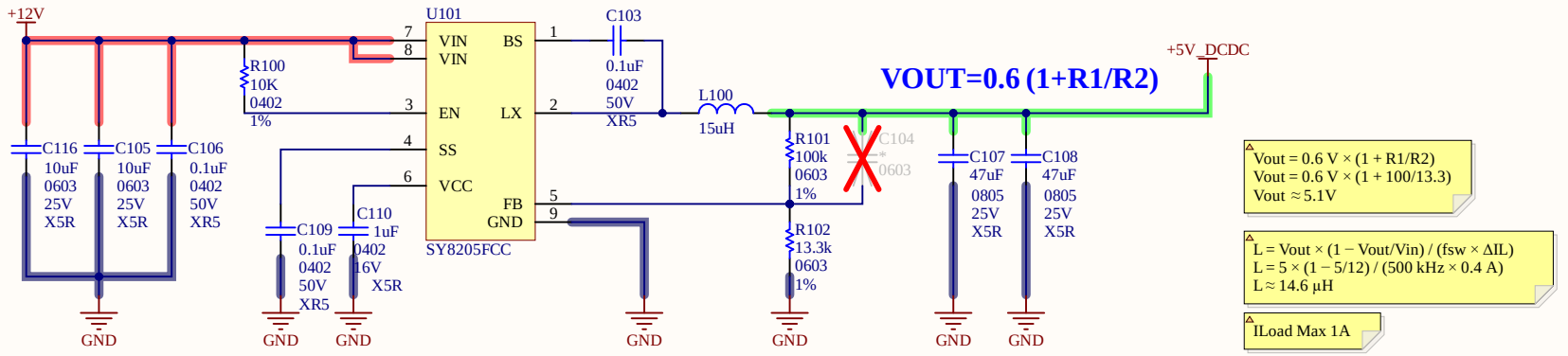
AC Input



EMI Filter

AC-DC 110V - 12V 1.6A

12V-5V 5A MAX

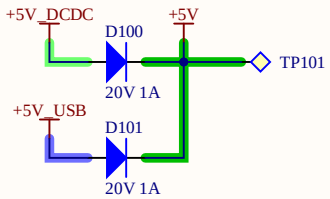


Vout = 0.6 V × (1 + R1/R2)
Vout = 0.6 V × (1 + 100/13.3)
Vout ≈ 5.1V

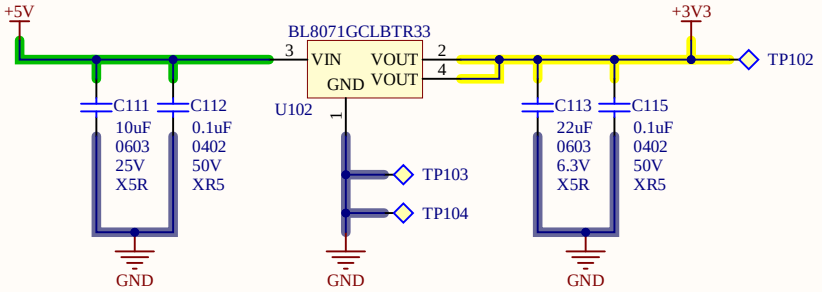
L = Vout × (1 − Vout/Vin) / (fsw × ΔIL)
L = 5 × (1 − 5/12) / (500 kHz × 0.4 A)
L ≈ 14.6 µH

ILoad Max 1A

5V Diode OR



LDO 3V3 1A



Title: **Power**			
Project: CABINET_IOT-V1.PrjPcb			
File: 100_Power Supply.SchDoc		Revision: V1R1	
Date: 20261112	Size: A4	Drawn by: J-E.Pachon-J.Jaramillo	www.somosinternet.com

STRAPPING PIN

MODE 2:1, 0|0|0 10Base-T Half Duplex. Auto-negotiation disabled.

MODE 2:1, 0|0|1 10Base-T Full Duplex. Auto-negotiation disabled.

MODE 2:1, 0|1|0 100Base-TX Half Duplex. Auto-negotiation disabled. CRS is active during Transmit & Receive

MODE 2:1, 0|1|1 100Base-TX Full Duplex. Auto-negotiation disabled. CRS is active during Receive.

MODE 2:1, 1|0|0 100Base-TX Half Duplex is advertised. Autonegotiation enabled. CRS is active during Transmit & Receive

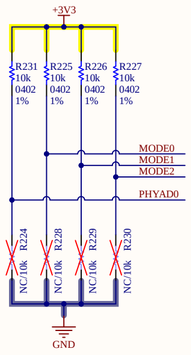
MODE 2:1, 1|0|1 Repeater mode. Auto-negotiation enabled. 100Base-TX Half Duplex is advertised. CRS is active during Receive

MODE 2:1, 1|1|0 Power Down mode.

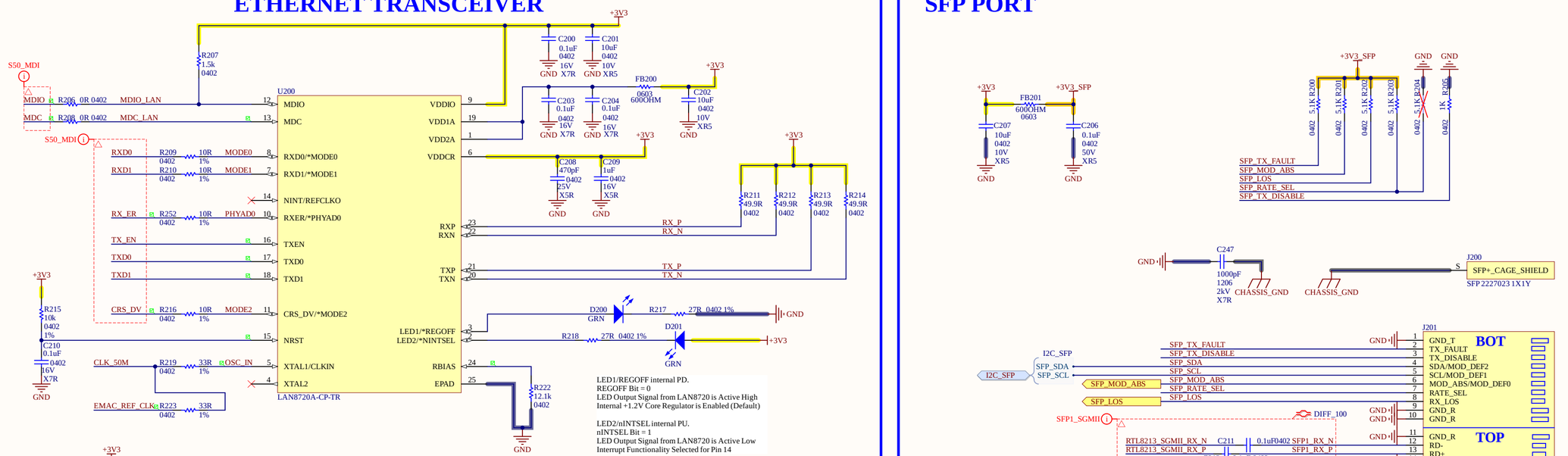
MODE 2:1, 1|1|1 All capable. Auto-negotiation enabled.

The PHYAD0 bit is driven high or low to give each PHY a unique address.

The device's SMI address may be configured using hardware configuration to either the value 0 or 1.



ETHERNET TRANSCEIVER



SFP PORT

STRAPPING PIN

DISAUTOLOAD/GPO45/P6LED1 =

-Pull Up: Disable EEPROM autoload
-Pull Down: Enable EEPROM autoload

MID29/GPO48/P6LED0 =

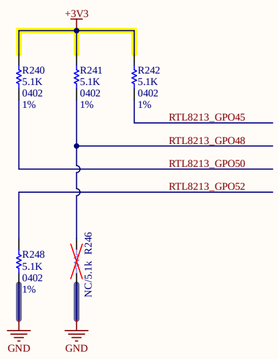
-Pull Up: Slave SMI (MDC/MDIO) Device Address is d'29 0x1D
-Pull Down: Slave SMI (MDC/MDIO) Device Address is 0

EN_PHY/GPO50/P3LED2 =

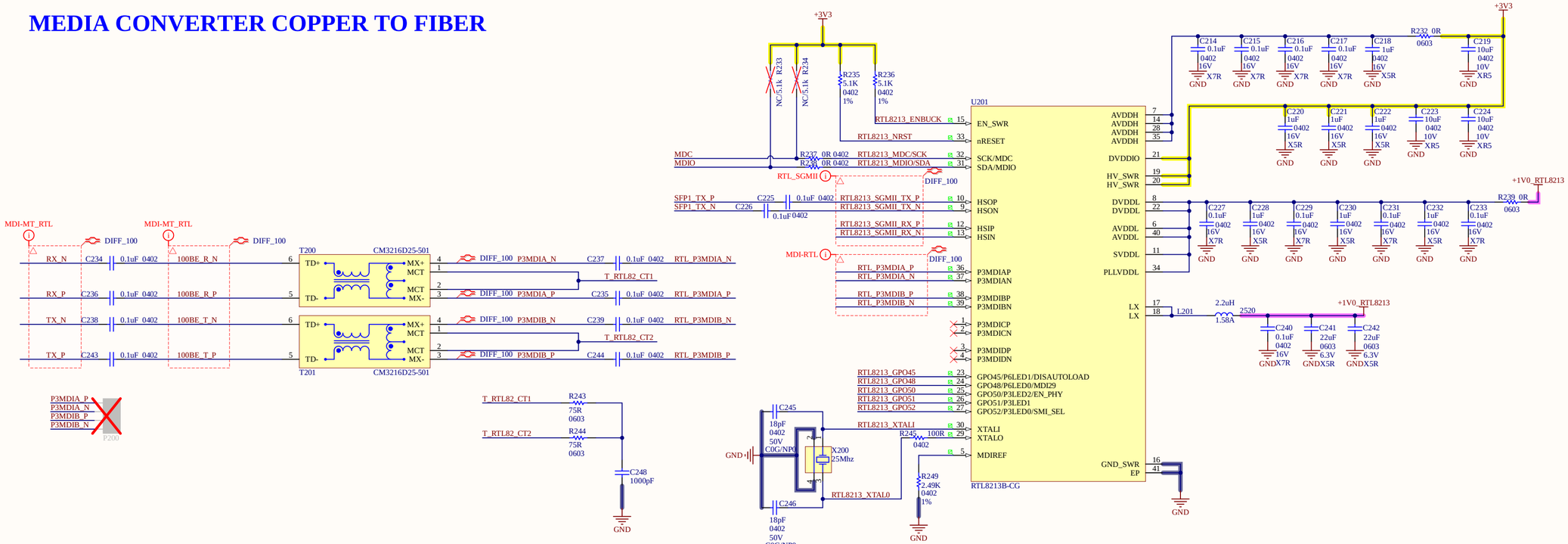
-Pull Up: Enable embedded PHY
-Pull Down: Disable embedded PHY

SMI_SEL/GPO52/P3LED0=

-Pull Up: Slave I2C-like interface
-Pull Down: Slave MII Management interface



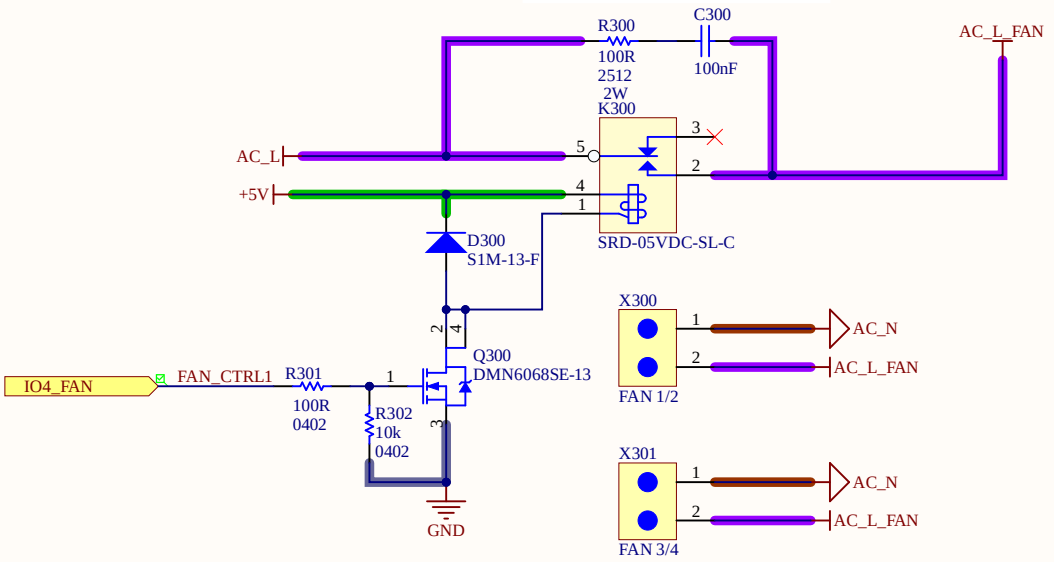
MEDIA CONVERTER COPPER TO FIBER



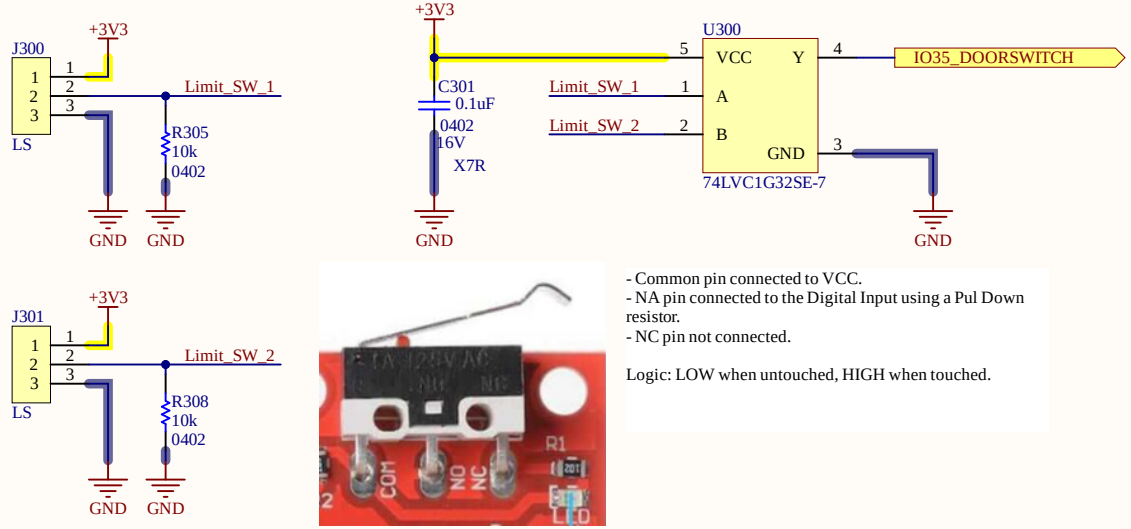
Title: **Ethernet**			
Project: CABINET_IOT-V1.PrjPcb			
File: 200_Ethernet.SchDoc		Revision: V1R1	
Date: 20261112	Size: A2	Drawn by: J-E.Pachon-J.Jaramillo	www.somosinternet.com

FAN CONTROL DRIVER

RC Snubber added for protection and EMI.



LIMIT SWITCH

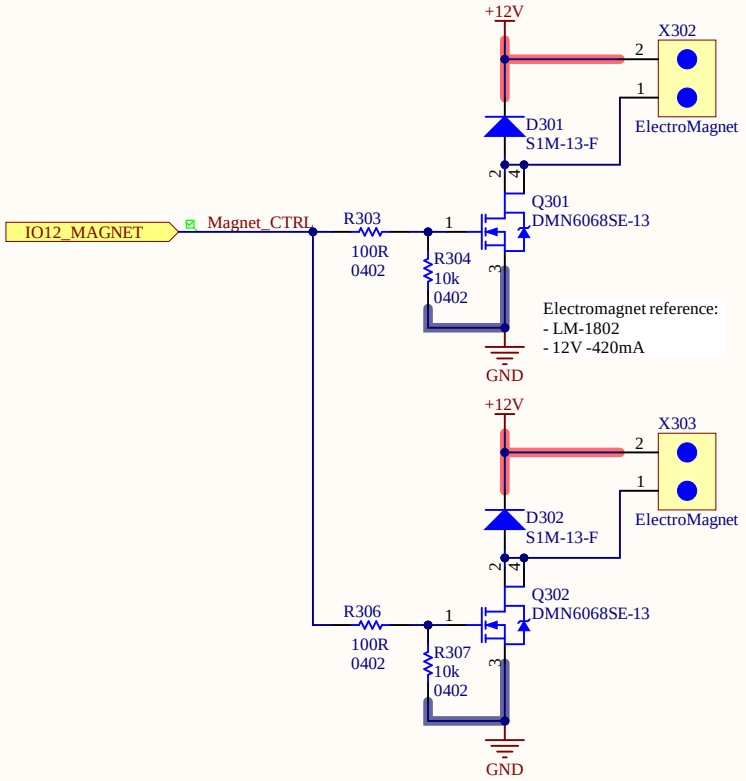


- Common pin connected to VCC.
- NA pin connected to the Digital Input using a Pul Down resistor.
- NC pin not connected.

Logic: LOW when untouched, HIGH when touched.

ELECTROMAGNET CONTROL DRIVER

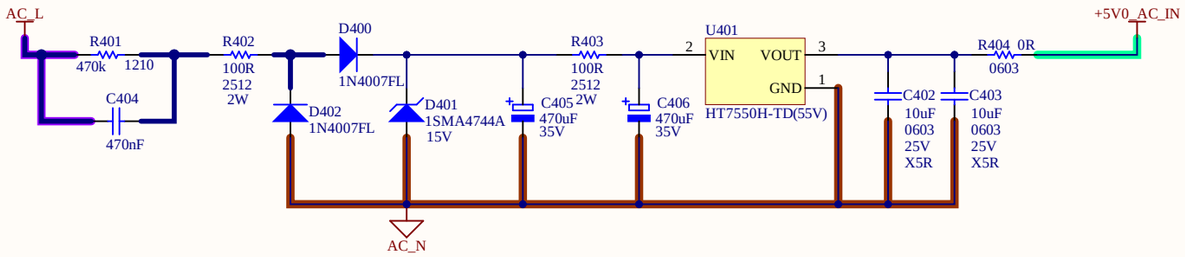
Modelo	LM1802	LM1805
Tensión de funcionamiento	DC12V	DC12V
Corriente de funcionamiento	420mA ± 5%	420mA ± 5%
Aumento de temperatura	+15℃ (Temperatura ambiente de 29℃; 6 horas después de que se energice DC 12V. La temperatura máxima del cuerpo de la cerradura es de 41℃).	
Interfaz	Rojo + Negro: -	Rojo: + Negro - Amarillo: NO Verde: COM Blanco: NC
Gama de fuerzas de tracción	120kg ± 8%	120kg ± 8%
Dimensiones (Ancho*Alto*Profundidad)	Cuerpo de la cerradura: 170*40*20(mm) Placa de hierro: 130*33*11(mm)	Cuerpo de la cerradura: 170*40*20(mm) Placa de hierro: 130*33*11(mm)
Peso bruto	1.016kg	1.029kg



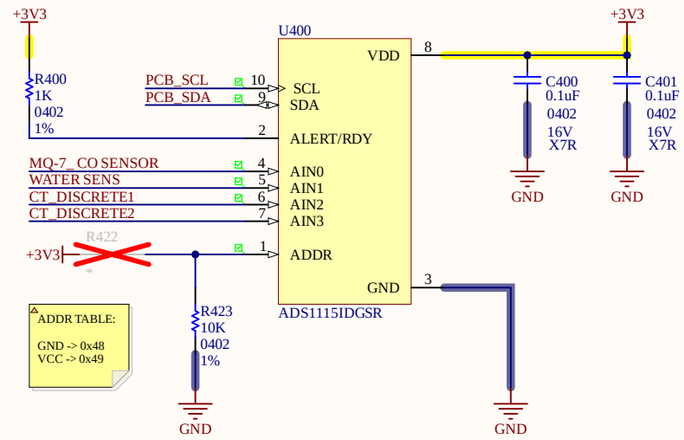
Title: **Control**			
Project: CABINET_IOT-V1.PrjPcb			
File: 300_Control.SchDoc		Revision: V1R1	
Date: 20261112	Size: A4	Drawn by: J-E.Pachon-J.Jaramillo	

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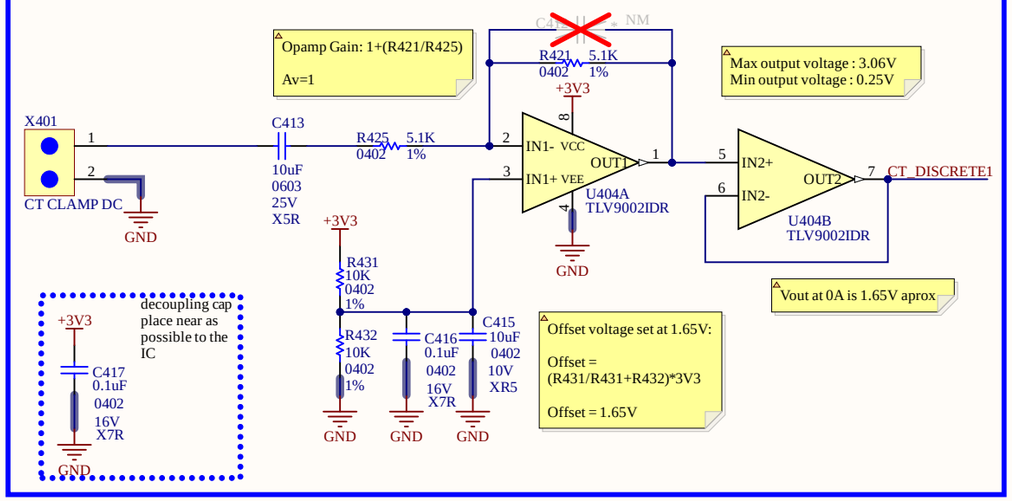
AC-DC CONVERTER STAGE 5V LDO



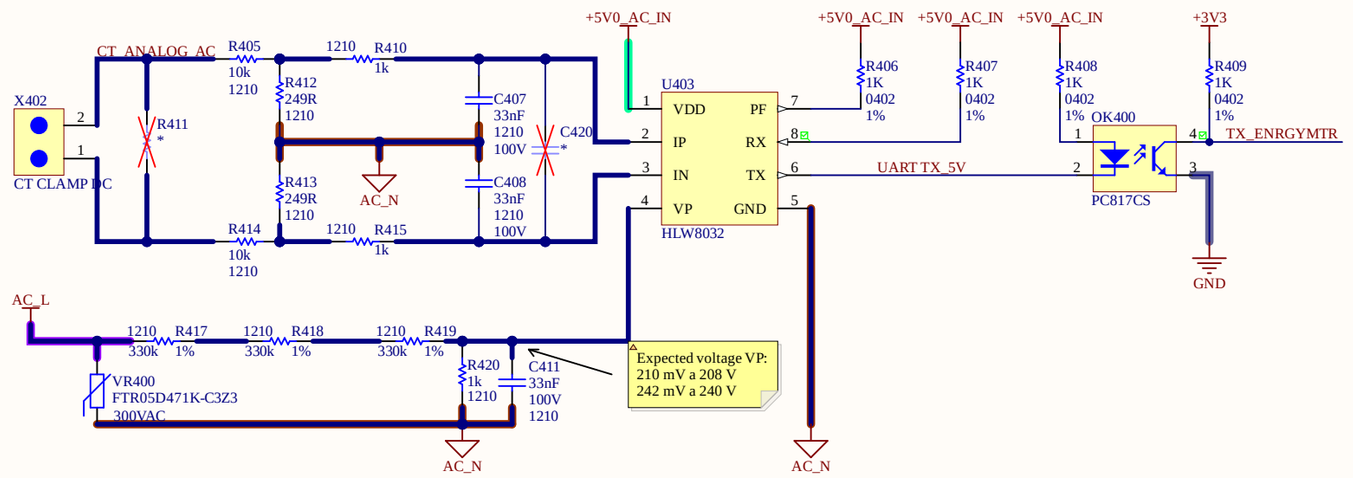
EXTERNAL ADC



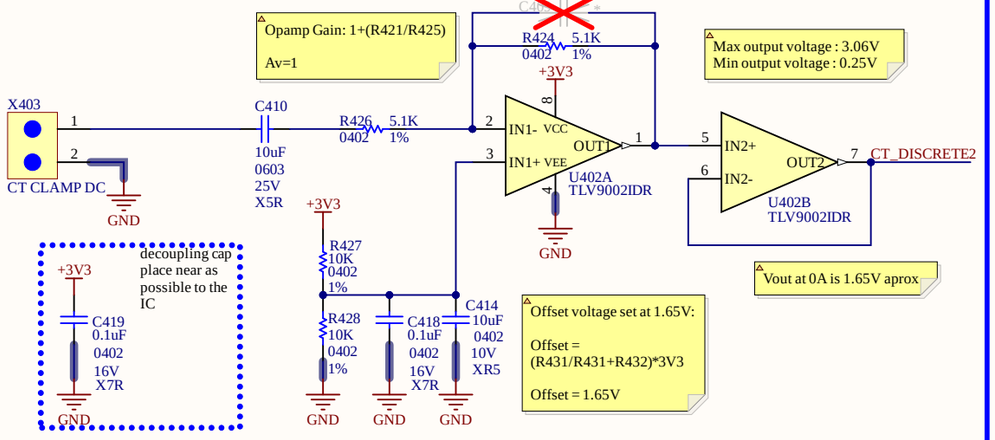
CT1 CONDITIONING CIRCUIT



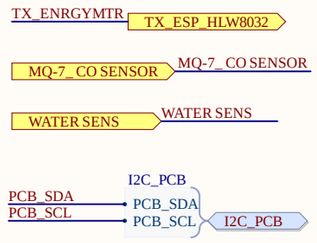
POWER METER



CT2 CONDITIONING CIRCUIT



PORTS & HARNESSES



ASSEMBLY NOTE:
MEASUREMENT OPTIONS

DEFAULT (A): HLW8032 IC (mounted by default).

OPTION B: ANALOG DC CLAMP MEASUREMENT
Mount: C413, C415, C416, C417, R425, R431, R432, U404, X401,
C410, R426, C409, R424, U402,R427,R428, C418, C414,C419,X403

Do not mount (NM): R410, X402

NOTA - MEDICION DE CORRIENTE CON CT (HLW8032)

CT: SCT-013-005, salida 1 Vrms @ 5 A (burden interna)
Sensibilidad CT: S_ct = 1 V / 5 A = 0.2 V/A

DIVISOR SIMETRICO
Rs = 10k 1% (x2, en terminales A y B del CT)
Rb/2 = 249R 1% (x2, punto medio a GND = N)
Rb = 2 x 249 = 498R
Rtot = 2*Rs + Rb = 20498R (carga sobre CT ~20.5k)

RELACION DEL DIVISOR
k = Rb / (2*Rs + Rb) = 498 / 20498 = 0.0243

VOLTAJE EN V1P-V1N
V1 = I * S_ct * k = I * 4.86 mV/A
@ 5 A: V1 = 24.3 mVrms (79% de FS)

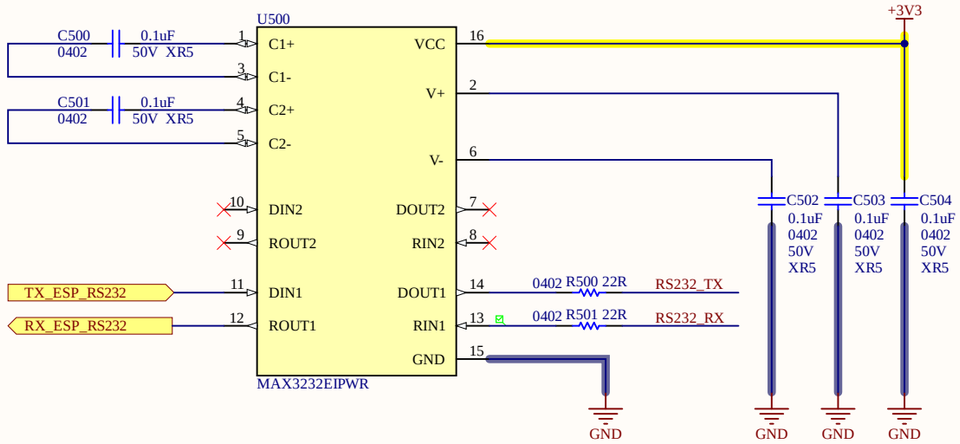
LIMITE HLW8032
V1_max = 30.9 mVrms (datasheet)
I_max = V1_max / (S_ct * k) = 6.36 A

SHUNT EQUIVALENTE (FIRMWARE)
R_eq = S_ct * k = 4.86 mOhm
Coef. corriente/potencia = 1 mOhm / R_eq = 0.206
(respecto al diseno de referencia con shunt 1 mOhm)
-> Calibrar con carga resistiva conocida

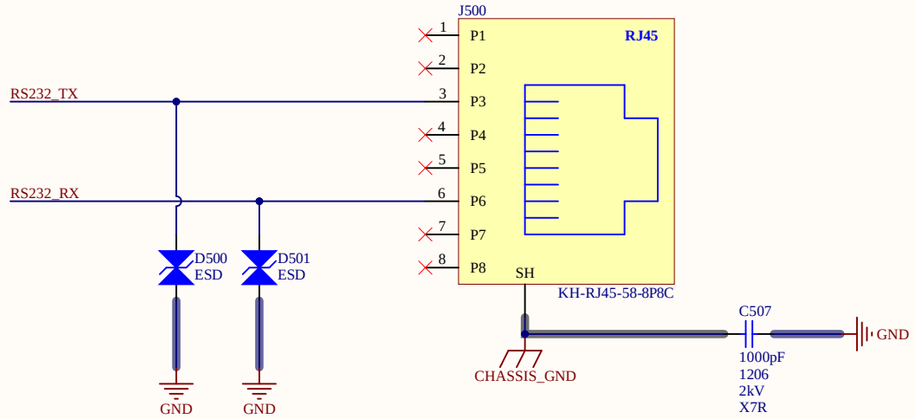
Title: *Sensors*			
Project: CABINET_IOT-V1.PrjPcb			
File: 400_Sensors.SchDoc		Revision: V1R1	
Date: 20261112	Size: A3	Drawn by: J-E.Pachon-J.Jaramillo	

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Converter UART to RS232

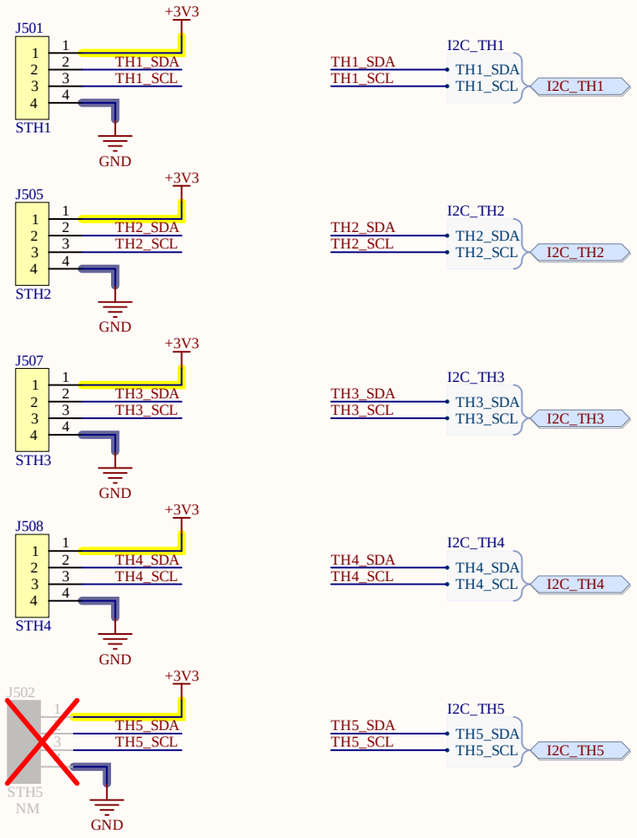


RS232 Connection with UPS



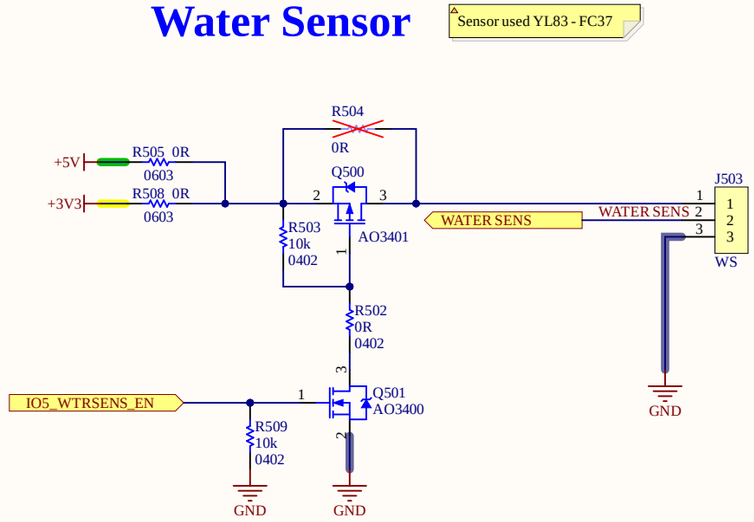
Temp & Hum

Sensor Used AHT20
Address 0x38



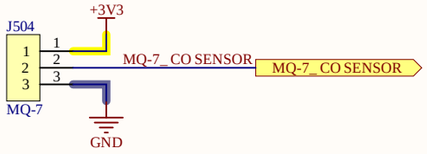
Water Sensor

Sensor used YL83 - FC37

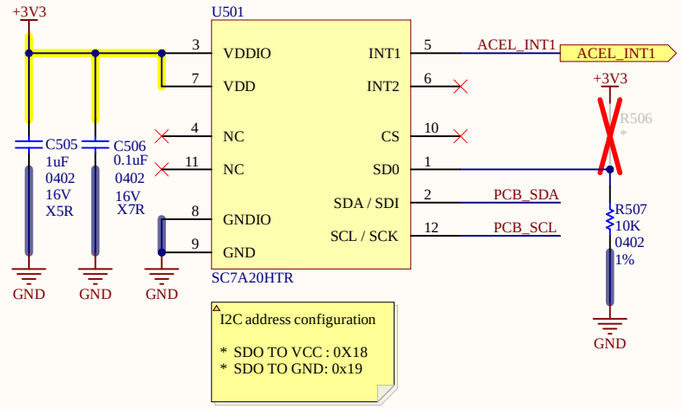


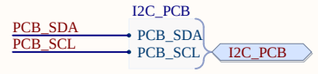
CO Sensor

Sensor CO used MQ7



ACCEL SENSOR





Title: External			
Project: CABINET_IOT-V1.PrjPcb			
File: 500_External.SchDoc			Revision: V1R1
Date: 20261112	Size: A3	Drawn by: J-E.Pachon-J.Jaramillo	

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